

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read: at the patch order of a simulated cortical sheet it constrains at most 10 sites, so on a larger population the positive link from coherence to content rests entirely on a bridge assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps

and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows consistency, not cortical realization. The conditions yield a disconfirmable comparison: decoders fitted independently on overlapping regions should agree more on reported than on matched unreported stimuli at equal phase coherence. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. No principle yet selects the cover, and the sheaf's overlaps are spatial where the example's are overlaps of content, so the proposed correlate of unity has no operational definition. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup's location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver's relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures

14 and fields; identifying them with perceptual content requires independent
15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which. Phase
23 order constrains shared content only through encoders related on overlaps,
24 with a bound that grows as the square root of the population it is read
25 on. Compatible local descriptions, not synchrony, determine a unique global
26 state. A contracting self-map fixes one state but can reconstruct accurately
27 only a family of bounded diameter. Each result names a measurement that
28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
30 Critiques of the temporal binding hypothesis argued that synchrony, where
31 it occurs, cannot say which features belong together or what they jointly
32 represent: that grouping has to be computed before synchrony could sig-
33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
35 and a pigeonhole count. What formalization adds to the informal argument is
36 quantitative and bookkeeping content. It gives the rate at which a coherence

bound on content weakens with population size and the population at which it stops constraining anything, the diameter within which one contracting map can reconstruct, and the number of codes a separated family of states requires. It also gives a complete inventory of the premises a synchrony-to-content argument consumes, each of which names a measurement, together with one model in which all of them hold at once.

The familiar noisy mean-field oscillator supplies a stationary coherence threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource condition for a declared actuator, bounds that connect phase order to content only through specified encoders, and conditions under which compatible descriptions glue and an accurate reconstruction fixes the glued state. A Lean 4 development checks these implications and a joint witness satisfies their eight connecting hypotheses in a specified model. It proves neither that cortex satisfies those hypotheses nor that the glued state is experienced. The latter identification motivates the framework and needs its own endpoint.

The empirical question is which measurements could support or disconfirm the proposed connections. A phase resultant can miss a locally ordered winding field; stationary phase distributions can leave the coupling mechanism unidentified; and coherent adaptation can lose alignment with a designated content template. Independent decoders must identify shared quantities before their agreement can be tested. We therefore distinguish stationary branches from finite observed onset, and compare prespecified field and connectivity onset models on held-out emergence recordings. The com-

60 parison tests a specified spatial prediction, not the presence of experience.
 61 Section 10.2 states the comparison that bears on conscious content directly.
 62 Sections 2–5 develop the three requirements; Section 6 applies their mea-
 63 surement obligations to a digital candidate. Section 7 states the resource and
 64 composition premises, and Sections 8–9 examine observables and the cortical
 65 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
 gives their derivations and controls.

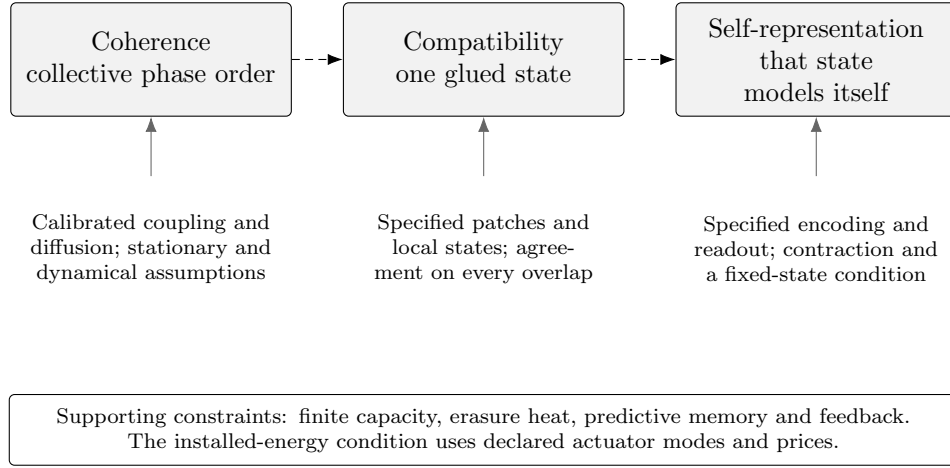


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

66

67 **2 Limits of phase order**

68 Several limits hold without the cortical hypothesis. Equal phases can coexist
69 with incompatible local descriptions; phase order constrains shared content
70 only through encoders related on overlaps. A winding field can have zero
71 global resultant while remaining locally ordered. A constant readout can
72 have a unique fixed point without reconstructing its input. A region that
73 does not distinguish relevant states cannot encode them, and a site cannot
74 report a change outside its causal past by a declared deadline. These lim-
75 its reflect familiar phase oscillator and information constraints (Acebrón et
76 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
77 The sections below state each one beside the condition or measurement it
78 motivates.

79 **3 Coherence: physical coupling and its re-** 80 **source condition**

81 **3.1 Dynamics and the stationary threshold**

82 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

83 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
84 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
85 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
86 quency heterogeneity and diffusion are distinct parameters. The stationary
87 calculation below uses identical frequencies and this scalar coupling. Ap-
88 plying it to a spatial kernel requires an aggregation rule and a check of the
89 resulting approximation.

90 For identical frequencies in their rotating frame, the positive normalized
91 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

92 where I_n are modified Bessel functions and ψ is the collective phase. The for-
93 mal derivation uses the absence of stationary current for a periodic gradient
94 drift to obtain this density, then verifies normalization and self-consistency.
95 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
96 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
97 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

98 Figure 2 shows this stationary branch.

99 Linearization about the uniform density gives rate $K/2 - D$ for the first
100 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists

101 exactly above threshold. About the coherent density, the quadratic form
 102 is strictly dissipative transverse to collective rotation, whose rate is zero.
 103 These linear estimates concern the limiting equation. Relating them to finite
 104 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 105 tion; exchange symmetry alone does not establish the needed independence
 106 (Sznitman, 1991).

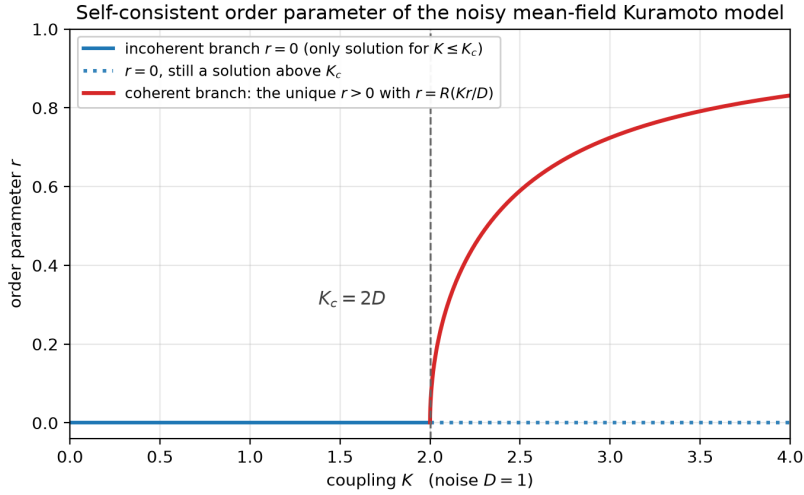


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

107 **3.2 Installed energy bounds mean coupling**

108 A spatial model replaces the finite sum by integration against a kernel. One
 109 declared implementation carries response modes with continuous profiles ϕ_i ,

110 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

111 The same configuration sets the reciprocal kernel and stored energy. An
 112 executed transition changes the kernel through its changed occupancies and
 113 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 114 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 115 pair reconstruction on refining spatial partitions converges to its product-
 116 measure integral under the specified compactness and sampling assumptions;
 117 the refinement index is independent of the number of control updates.

118 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 119 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 120 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

121 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 122 coherent stationary branch and any growing Fourier mode about the uniform
 123 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 124 is necessary: energy above it need not produce sufficient coupling because
 125 the bound can be loose.

126 The theorem concerns stored energy in the declared actuator modes. The
 127 estimated cortical signalling expenditure and extracellular electrostatic en-
 128 ergy in Supplemental Material measure different quantities; neither identifies
 129 the occupancies, mode prices or their relation to metabolic supply. Conse-
 130 quently the bound supplies a necessary condition for a declared actuator,
 131 but no cortical value of κ or demonstration that cortex meets it. Applying
 132 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 133 a justified reduction.

134 **4 Compatibility: from coordinated activity** 135 **to a global description**

136 **4.1 Why common timing is insufficient**

137 For overlapping patches U and V , compatibility means that restricting either
 138 local description to $U \cap V$ gives the same object. Descriptions can agree on
 139 shared quantities while contributing different information elsewhere. Com-
 140 patible sections of the specified sheaf have a unique global extension. On
 141 a finite cover, local finite spatial measures agreeing as measures on every
 142 overlap also have a unique finite-measure extension whose sheaf image is the
 143 glued section. No mass-one normalization is imposed.

144 In the cup example, visual and reaching descriptions must express a
 145 shared quantity, such as cup location, in common coordinates. Restriction

146 retains that quantity while dropping details particular to each description.
 147 Agreement permits surface appearance and hand configuration to contribute
 148 to one situation. The mathematical counterexample assigns the same phase
 149 to every patch while two local mass profiles disagree on a shared site. Those
 150 profiles have no common extension. Because they are chosen independently
 151 of phase, synchrony places no constraint on their disagreement.

152 This distinction limits the biological interpretation. Receptive territories
 153 and content domains must be related to spatial patches, and the descriptions
 154 must support the required extension. Matching overlap marginals of de-
 155 coded probability laws need not determine a unique global joint distribution
 156 (Abramsky & Brandenburger, 2011).

157 The cover is itself a choice, and the construction constrains it in one way
 158 only: synchronized patches agree where they overlap. Where they do not
 159 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 160 satisfies that obligation for *every* assignment of local descriptions and phases
 161 and the gluing theorem still returns a unique global section, which is the
 162 tuple it was handed. Membership of the class then reports nothing about the
 163 system beyond the patches covering it. What a gluing claim asks of a cover
 164 is therefore a non-degeneracy the theorem does not assume: that distinct
 165 patches meet, and that what they share is a quantity fixed independently of
 166 the agreement it is being used to establish.

167 The cup example exposes a second gap, between two kinds of overlap.
 168 The sheaf is built over a spatial substrate, so its patches overlap where they

169 share sites. Visual and motor territories that agree on cup location overlap
 170 in what they describe, not in where they lie, and the construction supplies
 171 no relation between the two. One route recasts the construction over content
 172 variables: each region becomes a partial decoder whose domain is the set of
 173 variables it estimates, and two regions overlap exactly where they estimate
 174 a common variable. The gluing argument needs only a cover and restriction
 175 maps, so it applies formally to such a cover, but no result here is stated
 176 for one, and which variables and decoders form it is an empirical question.
 177 Until a principle selects the cover, the proposed correlate of unity has no
 178 operational definition: gluing on a declared cover is proved, and which cover
 179 cortex realizes is open.

180 **4.2 When coherence does constrain content**

181 A declared relation between phase and content gives a positive result. Let
 182 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 183 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 184 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 185 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 186 sites; their equality on overlaps is a supplied relation between descriptions of
 187 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 188 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 189 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.

190 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (6)$$

191 Here discrepancy is the norm of the difference, and r and N are read on
192 whichever population the bound is applied to. That choice is what decides
193 whether the estimate says anything. A patch carries an order parameter
194 of its own, and reading the bound there carries that patch's resultant and
195 that patch's site count, so a large population observed through small patches
196 is constrained by the small count. Read instead on the whole array, the
197 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
198 from an average over N sites. The order cannot be improved in general:
199 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
200 observations agree exactly at either scale. A shared scalar encoder is a special
201 case. Dropping the overlap relation permits disagreement even at perfect
202 locking. A common nonexpansive linear projection preserves agreement, but
203 agreement of a projection does not establish agreement of the full vector.

204 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
205 constrains anything only where the right-hand side falls below that. At the
206 sheet's own patch order, the bound is informative for at most 10 sites; beyond
207 that, the coherence-to-content link requires the bridge assumption E78 of
208 the cortical hypothesis (Table 1 gives the full composition and its edges).
209 Supplemental Material provides the detailed derivation of this saturation,

210 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 211 by a count of disagreeing pairs, a chain across a cover's shared sites, or a
 212 spectral gap at a frequency-locked configuration.

213 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 214 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 215 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 216 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

217 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 218 duce an initial mismatch. Approximate observations leave a nondecaying
 219 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 220 update to the phase process.

221 4.3 What approximate agreement determines

222 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 223 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 224 Weights that vanish outside their patches and sum to one at each site select
 225 a global profile by weighted averaging. The selection lies within ε of each
 226 patch there, and any two such selections differ by at most ε , independently
 227 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 228 $L\varepsilon$ across selections.

229 At zero tolerance, the selection is the unique exact extension. At every

230 positive tolerance there are families with no exact extension; selection then
 231 gives an approximation whose dependence on weights is bounded. Supple-
 232 mental Material gives a three-site witness attaining the bound. The mass-
 233 profile construction uses the scalar case of the content dynamics and addi-
 234 tionally requires nonnegativity of the initial profiles and observations. The
 235 general vector result supplies no identification with the spatial-measure sheaf.

236 These constructions concern declared objects: a specified cover, contents
 237 valued in a normed space, and encoders related on overlaps. They select
 238 neither the neural cover nor the decoder, and identifying their contents with
 239 perceptual content, or establishing that the identification bears on experi-
 240 ence, is a further modelling task; Section 8 sets out what a measurement of
 241 it would have to establish.

242 **5 Self-representation: a state that models it-** 243 **self**

244 **5.1 The reflexive construction**

245 Compatibility gives a global state. Reflexivity relates it to an encoding within
 246 that state. Equip global sections with a complete metric. If the additional
 247 restriction-resonance condition identifies encoding with restriction to a region

248 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

249 The region and readout are fixed across the state space. If F contracts and
 250 fixes the section s_* formed from the local descriptions, Banach’s theorem
 251 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 252 (Table 1 gives the full composition and its edges) supplies the contraction
 253 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
 254 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
 255 a relaxation time $\tau > 0$. Constructing a map with that factor remains a
 256 modelling obligation. A fixed contraction can reconstruct accurately only a
 257 restricted family of states: if its error is at most ε throughout that family,
 258 any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

259 In the cup example, the global state includes the represented scene and the
 260 perceiver’s relation to it. A local encoding must retain enough information
 261 for the readout to recover that modelled state; cup position alone would be
 262 insufficient if hand configuration and viewpoint could not be reconstructed.

263 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 264 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

265 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 266 fixed positive ε and τ . Exact uniqueness survives while the error guarantee

267 weakens.

268 5.2 Reconstruction across a family of states

269 Sensitivity can be required quantitatively. Declare relevant macrostates, an
270 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
271 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

272 Thus M relevant states pairwise more than 2ε apart require at least M
273 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
274 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
275 existing self-prediction map.

276 This is an additional accuracy criterion on a family, with no requirement
277 for multiple exact fixed points. For the same fixed contracting map used by
278 E89, however, the diameter bound above is an additional restriction: at zero
279 error the family contains at most one metric-distinct state. A broad family
280 therefore requires a different, possibly input-conditioned reconstruction map
281 with its own accuracy and stability assumptions. A blind encoder or constant
282 readout fails the stated criterion for separated states despite having a unique
283 fixed point.

284 Where the encoding is read as the state's own restriction to a region, the
285 criterion becomes a condition on the region. Restriction resonance requires

286 the encoder and that restriction to identify the same pairs of states, so a
 287 boundary reconstructing two relevant states more than 2ε apart while its
 288 region reads them identically is not resonant. Read forwards, a resonant
 289 boundary that reconstructs a pairwise separated family restricts injectively
 290 on it. Which states a region reads identically is fixed by where the region
 291 sits, before an encoding is chosen, so the resolution the family demands is
 292 the region's and no readout supplies it. The implication runs one way: a
 293 region that does resolve the family is not thereby shown to carry a resonant
 294 encoding, and E89 supplies no such resolution.

295 Resonance and resolution are jointly satisfiable without a local fold. Take
 296 the region to be the whole substrate and use the identity readout: restric-
 297 tion is the identity, so this boundary is resonant and reconstructs every de-
 298 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 299 E89's strict contraction premise on a nontrivial metric space. A properness
 300 condition on the region is therefore a separate requirement, and the fixed-
 301 contraction diameter bound remains in force when E89 is also required. With
 302 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 303 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 304 so the family is no larger than the δ -packing number of what the region car-
 305 ries. That is the shape a continuous family of states needs, and at the whole
 306 substrate it says nothing, because the region's range is everything. The gain
 307 L is declared and derived here for no physical readout, and at unbounded
 308 gain the bound says nothing.

309 **5.3 Agreement after a change takes communication**

310 Snapshot compatibility does not specify how quickly descriptions respond
311 when a shared quantity changes. In a synchronous network whose updates
312 read only the site’s own state and declared neighbours’ messages, two initial
313 configurations agreeing on a site’s causal past through T rounds produce the
314 same state there at round T . The deadline consequence is the one Section 2
315 states, and it follows from the reconstruction bound in Section 5.2: a report
316 from that site is a readout of a code that the two interventions leave identical.

317 A three-site delay line witnesses failure after one round and exact report-
318 ing after two; cutting the path prevents delivery at every round. A report
319 may still happen to be correct about one fixed world. The bound concerns
320 responsiveness across interventions.

321 Throughout this section *prediction* names a reconstruction map, and the
322 relevant family of states, the metric, the tolerance, the communication graph
323 and the deadline are all declared inputs, as are the represented variables and
324 the fidelity counted as accurate. Reading the map as temporal neural pre-
325 diction would require dynamics; reading the resulting code count as physical
326 memory capacity would require a hardware model, which a state-dependent
327 readout would have to extend.

328 6 What the conditions constrain in a digital 329 candidate

330 The conditions bear on a large language model (LLM) running on graphics
331 processing units (GPUs) only through measurements, and only for a declared
332 task. For a finite spatial coupling operator with eigenvalues $|\lambda_1| \geq \dots \geq |\lambda_N|$,
333 any linear encoder and readout factoring through d real coordinates pays re-
334 construction error at least $\sum_{i>d} \lambda_i^2$. That rank premise is a property of an
335 explicitly linear bottleneck; query/key dimension does not supply it for soft-
336 max attention. On one synthetic two-view task whose target is the glued
337 state and a self-relative coordinate, a rank-two linear decoder has predicted
338 training-error floor 2.000 and held-out error 2.000 on a parity-held-out family.
339 The deadline premise is similarly specific: within one forward pass, the causal
340 mask confines position t to its causal past, so an intervention outside it can-
341 not be reconstructed there by that step. These results rank no architecture
342 family, imply no general inferiority of GPU hardware, and settle no question
343 of machine experience; a claim of unity or self-representation for such a sys-
344 tem commits its maker to identifying the shared quantities, the code and its
345 separation of the declared family under intervention. Supplemental Material
346 gives the bottleneck derivation, the task controls, the information ceiling of
347 token outputs and the hardware scope.

348 7 Resource premises and composition

349 Two thermodynamic statements enter, with different standing. The first is
350 proved for the phase model: for a smooth positive periodic density with prob-
351 ability current J and diffusion D , the current functional $\sigma_J = \int J^2/(D\rho) d\theta$
352 bounds how fast order can change, $\int_0^\tau \sigma_J dt \geq [\arcsin r(\tau) - \arcsin r(0)]^2/(D\tau)$
353 while $0 < r < 1$, and stationary order has no positive floor. The second is
354 conditional: the installed-energy condition of Section 3.2 is necessary for its
355 declared actuator, and restricts a physical candidate only once mode prices,
356 coupling reduction and diffusion are measured. Neither is a heat or power
357 statement until the current is identified with a physical flux, a reservoir at a
358 stated temperature is specified and local detailed balance holds for the steps
359 concerned; only then does σ_J convert to entropy production and the finite
360 register budgets to Landauer heat (Landauer, 1961; Norton, 2005; Shenker,
361 2000). No composition edge derives coherence, compatibility or reconstruc-
362 tion from a heat bound. Finite-register capacity obeys $H(p) \leq \log |X|$, and
363 a separate boundary result concerns a field reaching disconnected vacuum
364 components (Kibble, 1976); Table 1 states the connecting assumptions. Sup-
365 plemental Material gives the current-cost derivation, predictive-memory and
366 feedback balances, learning and repeated-operation budgets, and the sensi-
367 tivity of the metabolic comparison to its assumptions.

368 7.1 What the composition establishes

369 The passive and feedback composition theorems each accept eight connecting
 370 implications (Table 1). In the active branch, a named substep at the register’s
 371 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

372 with B the register’s heat. The upper allocation follows from a ledger whose
 373 operations deliver that total heat to one reservoir and whose other opera-
 374 tions do not increase joint entropy. Those ledger assumptions are physical
 375 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
 376 plies them. Finite two-bit witnesses exhibit positive forward and reverse
 377 elementary paths, local detailed balance at one thermal scale, and interac-
 378 tion energies of order $k_B T$; these conditions alone do not quantify proximity
 379 to equilibrium. Cortex has not been shown to satisfy these premises. The
 380 illustrative cortical metabolic expenditure over one region and one phase-
 381 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
 382 with the register heat budget B requires a common physical ledger. Read-
 383 ing the register at the scale where its premises are physical, such as a single
 384 channel, requires E45Active to carry that step’s budget to the convergence
 385 of a continuum kernel. The active branch is an alternative route to the same
 386 node rather than a stronger one, and no result of Section 3.2 passes through
 387 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical	A finite-capacity system realizes a field
<i>capacity and</i>	premise	leaving its vacuum manifold. Capacity and
<i>boundary</i>		boundary are independent roots.
E23	Bridge	The boundary is associated with a finite
<i>boundary to</i>	assumption	register whose update leaves states
<i>register</i>		unreachable.
E34 /	Bridge	Passive: predictive memory at the register’s
E34Active	assumption	thermal scale and heat budget, with
<i>register to</i>		nonzero waste. Active: positive paths, local
<i>dissipation</i>		detailed balance and allocation of register
		heat to the named substep; the allocation
		uses the ledger identity and other
		operations’ compressiveness. Both are
		exhibited at thermal-scale heats on finite
		state spaces with positive reverse paths,
		which is not the regime a cortical register
		occupies.
E45 /	Modelling	Convergent spatial coupling energies.
E45Active	assumption	Declared mode hardware supplies a kernel
<i>couplings to</i>		limit for either branch. A learning witness
<i>continuum</i>		computes installation and work from its
		own executed law. Convergence follows
		from mesh refinement rather than from the
		budget, which selects no regime at
		macroscopic scale. Refinement and physical
		identifications remain inputs.
E56	Physical	A specified physical field realizes the
<i>field</i>	commitment	limiting coupling and satisfies the kernel
<i>identification</i>		conditions.
E67	Physical	That field operates above the
<i>above</i>	commitment	synchronization threshold.
<i>threshold</i>		

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

388 Both branches also take a thermodynamic cover carrying exact over-
389 lap compatibility and a reflexive boundary specifying encoding and read-
390 out. They conclude that the glued section is the unique fixed point of self-

391 prediction. Compatibility and accuracy at that section enter as premises;
392 uniqueness follows. Identifying encoding with restriction needs the separate
393 resonance condition of Eq. (8).

394 Each of the eight edges is supplied rather than derived from the one above
395 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
396 pendent roots rather than a step from one to the other; both E45 discharges
397 take their convergence from refinement data and not from the thermody-
398 namic bound; E78 states that coherence relaxes the supplied cover, and the
399 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
400 coders already shared on overlaps. The inferential content of the composition
401 is its last step, where contraction and a fixed glued section give uniqueness.
402 What composing them establishes is that the premises are consistent, named
403 in one place and machine-checked together, and not that coherence follows
404 from installed energy or self-representation from coherence.

405 Joint witnesses combine a register, boundary, refining mesh and small
406 spatial cover; active witnesses add controlled thermal updates and a priced
407 installation. They establish satisfiability under their stated identifications.
408 Mesh convergence follows from refinement data, and the witnesses establish
409 no common cortical mechanism. Supplemental Material maps the claims to
410 their Lean declarations and records the counterexamples. The development
411 declares no additional axioms beyond its libraries' foundational axioms.

412 8 What the observations establish

413 The composition holds under its premises. Whether observation can estab-
414 lish those premises is a separate question, and the three requirements it
415 connects concern different objects: phase dynamics, local descriptions and
416 internal encodings. Their observables must retain these distinctions. The
417 following controls show which mechanisms remain indistinguishable, which
418 temporal approximations fail, which states one observable cannot separate,
419 and what independent content measurements must establish. Each is a finite
420 numerical regime on a declared model, with full parameters and controls in
421 Supplemental Material, and none of them calibrates a cortical quantity.

422 8.1 Stationary distributions leave the mechanism un- 423 determined

424 Equation (2) gives a concentration–coherence relation with no fitted shape
425 parameter. Estimate r from circular resultant length and a from density
426 shape: inverting R from the measured r would make agreement automatic.
427 Even independently estimated agreement identifies a statistical family shared
428 by different mechanisms.

429 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
430 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
431 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
432 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

433 and diffusion D_0 have that same density. Their stationary distributions co-
 434 incide window by window despite different causes.

435 Resolved dynamics supply additional information. In the ideal model
 436 with continuous latent paths, quadratic variation accumulates at rate $2D$
 437 and conditional increment means constrain drift; a filtered observation of
 438 the phase needs its own observation model before either is read off. Separat-
 439 ing endogenous coupling from a matched common drive additionally requires
 440 a measured drive or controlled perturbation; identifying electromagnetic cou-
 441 pling requires evidence specific to that interaction. Supplemental Material
 442 gives the equivalence construction and its numerical check.

443 8.2 Finite ramps change the observable onset

444 The square-root law in Eq. (3) describes stationary solutions. A changing
 445 coupling requires time for the phase distribution to approach those solutions.
 446 In a deterministic mean-field control that crosses the threshold symmetrically,
 447 none of the scanned rates meets the declared tracking tolerance: slowing
 448 the crossing does not remove the terminal residual under that initialization.
 449 A slow control variable therefore supplies no general guarantee of branch
 450 tracking at onset. Supplemental Material gives the residual, rate range and
 451 admissible rates away from threshold.

452 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 453 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 454 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay

455 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 456 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 457 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 458 onset exponents differ from those of the exact stationary branch evaluated
 459 on the same grids. At the fastest speed the ensemble mean never reaches the
 460 fit window, so its fitted slope does not estimate stationary onset.

461 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 462 makes the apparent exponent depend on ramp speed and fitting window.
 463 The specified deviation-based critical ramp speed is not bracketed by this
 464 sweep. With fluctuating coupling, the relevant stationary comparison also
 465 depends on rate: the mean coupling approximates order in the tested fast
 466 regime, while averaging stationary order over coupling works for the tested
 467 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 468 neither approximation.

469 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 470 $v^{0.444}$. The interval from fitting those means is [0.394, 0.494]; resampling
 471 replicas independently within each speed gives a 95% interval [0.398, 0.496],
 472 with fraction 0.019 of 10000 bootstrap fits at or above 1/2. This fraction
 473 describes the resampling distribution under the fitted protocol; it is not a
 474 test of the ideal exponent. A deterministic mean-field run seeded at the
 475 fluctuation floor gives exponent 0.425 with the same estimator and escape
 476 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls

477 show that the criterion affects the fitted exponent, but do not prove that it
 478 accounts for the entire finite-population shortfall. The larger-population low-
 479 threshold fit has substantial log residuals; matched-speed fits do not converge
 480 toward one-half across the tested sizes and criteria. Some trajectories also
 481 cross the absolute escape level before K_c , so the first postcritical crossing can
 482 yield a zero or grid-limited delay. This finite sweep establishes no common
 483 asymptotic exponent.

484 The ideal delayed-onset calculation motivates a square-root scaling hy-
 485 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 486 tify the coupling medium. A within-subject emergence design can vary the
 487 propofol taper and test how an operational onset changes with that manipu-
 488 lation, but interpreting the result as a coupling-rate law requires a measured
 489 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 490 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 491 describes how D could be estimated from resolved phase traces.

492 Additional controls address integration bias, population size and spatial
 493 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 494 $[1.958, 2.009]$ contains 2.

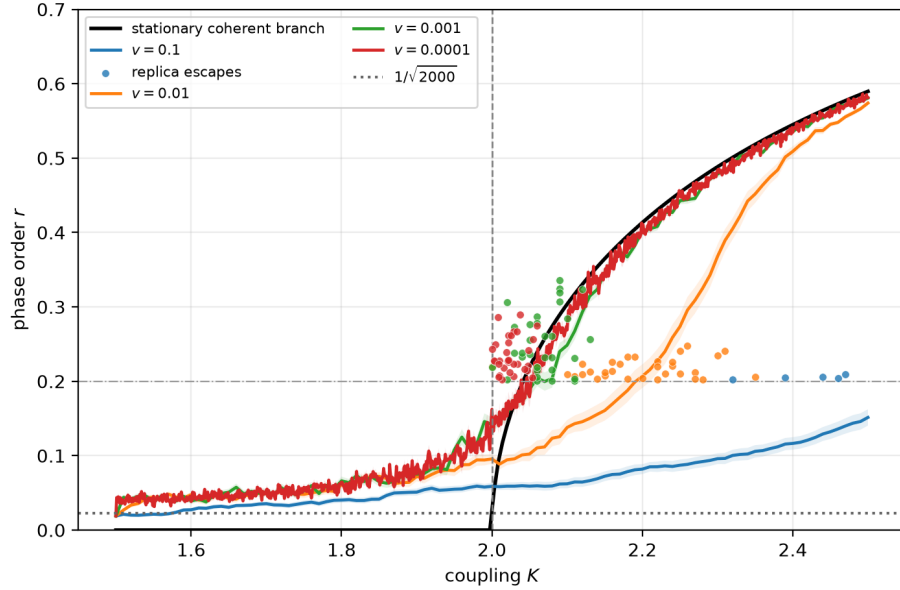


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica’s first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

495 8.3 A winding state leaves the order parameter am- 496 biguous

497 The global order parameter r discards spatial arrangement. On a periodic
498 sheet, a phase field can wind once around the sheet and have $r = 0$ while
499 neighbouring sites remain locked. At identical frequencies, every such wind-
500 ing is stationary for an isotropic symmetric kernel: contributions at opposite
501 separations cancel. It violates the quarter-turn initial-spread premise of the

convergence result, which is why that result can still select the uniform state.

In the spatial control, the winding persists below the interpolated decay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform and winding starts under the same coupling, diffusion and kernel end at $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched frequency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm, and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen realizations. These are finite controls, not a cortical wave identification or a derived transition.

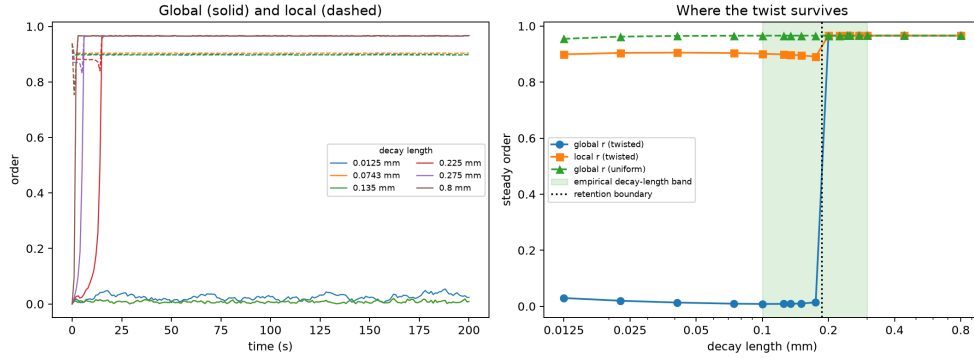


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

Linearization about a winding reweights each separation’s coupling by

512 the cosine of its phase advance. The resulting symmetric kernel has explicit
 513 mode rates: nonnegative reweighted coupling gives no growing mode, while
 514 a positive rate destabilizes that winding. The criterion concerns linear rates,
 515 not asymptotic stability, and the measured retention boundary need not
 516 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 517 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 518 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 519 winding from that model even though the phase-only equations retain it as
 520 a stationary state.

521 Neither model turns these states into measured cortical spirals. The pe-
 522 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 523 no propagation speed, and the amplitude variable has no calibrated relation
 524 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 525 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 526 winding patch when shared encoders are supplied, although its global $r = 0$
 527 form is uninformative. A nonzero ring winding number q also obstructs a
 528 global real-valued phase lift; it has no identified content meaning here. Thus
 529 an onset criterion based only on r can register a change of basin as well
 530 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 531 observable.

532 8.4 Coherent adaptation can lose environmental align- 533 ment

534 An adaptive coupling rule might supply representational structure, but its
535 objective must reward the relevant agreement. For symmetric coupling, pair-
536 wise sine terms cancel in the population mean drift. Minimizing squared drift
537 at that fixed mean therefore rewards equalizing drift across nodes. In the
538 frequency-cluster control, this favors coupling between unlike-frequency clus-
539 ters, whereas the diagnostic environmental template rewards coupling within
540 clusters.

541 The formalization proves descent of the squared-drift objective for a spec-
542 ified gradient flow at fixed phases. Its functional differs from thermodynamic
543 entropy production: at a coherent stationary density the probability current
544 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
545 rial gives the operator, its gradient and the conditions of the descent result.

546 The finite control co-evolves noisy phases and symmetric, nonnegative
547 couplings at fixed total resource. Its template is a diagnostic covariance
548 of latent signals, never a training input. Across three illustrative seeds,
549 second-half order remains between 0.9167 and 0.9222, while the within-
550 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
551 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
552 trols are reported in Supplemental Material.

553 The immediate post-update objective reduction disappears when aver-

554 aged over complete update intervals. A prespecified grid of 6 rate–cadence
555 combinations contains no case meeting the 5% sustained-reduction criterion
556 while retaining coherence and losing alignment. The finding is coherent adap-
557 tation away from the specified structure, with no demonstrated sustained cost
558 reduction.

559 Kernel alignment and overlap compatibility are different observables.
560 This control addresses acquisition of its designated template under its par-
561 ticular objective. Learning models can explicitly include data in an objective
562 (Corberi et al., 2026); an account of agreement about the cup must specify
563 the content its learning rule is meant to preserve.

564 8.5 Content agreement needs independent decoders

565 A neural test needs independently identified domains, local descriptions and
566 restriction maps. Two decoders should estimate a shared content on over-
567 lapping domains before their agreement is examined. Discrepancy requires a
568 null under decoding noise and controls using deliberately compatible and in-
569 compatible examples. An implementation on synthetic data separates these
570 cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
571 the compatible construction scores 0.476 on an interval containing one half.
572 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
573 bias can manufacture disagreement. Per-patch decoding diagnostics detect
574 these failures; the compatibility statistic alone does not.

575 Phase agreement cannot replace that measurement. On constructed con-

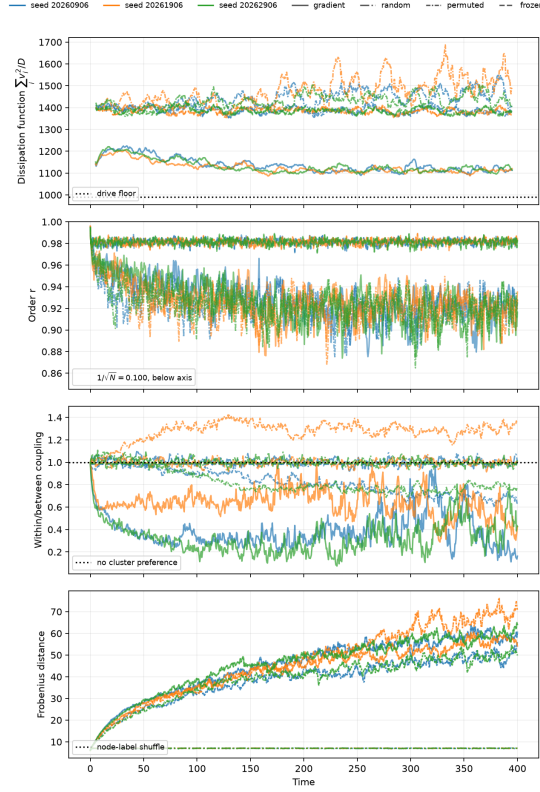


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

576 tent, phase carries a share 1.000 of the content entropy when content is built
577 from it and 0.000 when it is not. A phase-derived statistic reports com-
578 patibility on phase-locked patches whose local descriptions disagree. The
579 implemented content statistic detects their incompatibility. These controls
580 validate a measurement on known constructions; a neural content model and
581 decoder remain to be established.

582 Choosing the cover also chooses a candidate subject: trivial descriptions
583 can agree while carrying little content, and a partition can conceal disagree-
584 ment. The useful choice must therefore be constrained by what neural ac-
585 tivity encodes. The exact composition assumes compatibility of its supplied
586 cover. The content mechanisms above identify additional hypotheses under
587 which agreement can arise; they leave the empirical choice of content and
588 the connection to that cover explicit.

589 **8.6 Estimator calibration on exploratory EEG**

590 The concentration range and the dependence structure decide whether the
591 concentration-coherence relation can be tested at all. Under the specified
592 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
593 requires 3000 independent sites in the scanned designs, rising to 31000 when
594 sites share a mean in groups of 10. These counts describe that estimator and
595 dependence model at a declared design concentration; the intended recording
596 requires its own calibration. Supplemental Material reports the bin-count
597 controls and the full dependence-aware design.

598 The estimators were also exercised on public propofol-sedation EEG (Ba-
 599 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
 600 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
 601 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
 602 the residual from the Bessel relation is small. That agreement is uninforma-
 603 tive. Pooling over a window does not remove the common rotation of the
 604 phases, so a perfectly phase-locked 10-Hz signal passed through the same
 605 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
 606 instantaneous spatial resultant is one. The bipolar montage also removes
 607 spatial coherence: a field whose raw channels have instantaneous resultant
 608 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform
 609 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
 610 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
 611 the estimator’s behaviour and establishes no spatial concentration, coher-
 612 ence level or test of the stationary density. Sedation recordings also leave the
 613 awakening hypothesis untested.

614 An informative dataset therefore needs an estimand that removes the
 615 common rotation, a montage whose effect on spatial phase is modelled, suf-
 616 ficient concentration range or precision, calibrated estimators and an ob-
 617 servation model for dependence and source mixing. Intracranial recordings
 618 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
 619 informative concentration range, but that range must be measured. The
 620 design counts above are calculations for the specified estimator, not sam-

621 ple requirements transferable to another recording. Public mouse calcium
622 imaging provides thousands of spatially located neurons across waking, sleep
623 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
624 phase-estimation band used here. Site count, state coverage and bandwidth
625 are separate requirements.

626 **9 The cortical hypothesis and an awakening**

627 **test**

628 The cortical proposal identifies a physical source of coupling and asks whether
629 its changes help explain recovery. The tests in Section 8 determine what
630 observations must distinguish before a phase trace can support this identifi-
631 cation.

632 **9.1 Physical identification and calibration**

633 The cortical hypothesis identifies part of effective coupling with endogenous
634 extracellular electromagnetic interactions. Fields can modulate spike timing
635 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
636 propose coordination of distributed neural information (Pinotsis & Miller,
637 2022; McFadden, 2020). Phase organization under propofol motivates study-
638 ing this regime (Bastos et al., 2021), but does not measure coupling.

639 The formal field predicate requires a jointly continuous kernel, a normal-

640 ized substrate, coupling equal to the kernel average, and no single site pro-
641 viding more than half that strength. Toy constructions satisfy it. A cortical
642 realization must specify the scale at which the kernel and its reduction ap-
643 ply: calcium imaging reports different state-dependent modularity patterns
644 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

645 Cortical columns are the obvious candidate unit, and they answer one
646 of the two demands the formalism makes. As the oscillator index they are
647 a serious proposal: they are delineated by response properties and anatomy
648 rather than by the agreement they would be used to establish (Mountcastle,
649 1997), and a phase can be estimated from a column-sized territory at the
650 electrode spacings used for cortical wave measurements (Townsend et al.,
651 2015). As the content cover they are a poor one, for three separate reasons.
652 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
653 the compatibility obligation without constraining anything (Section 4). The
654 cover condition additionally requires the patches to exhaust the substrate,
655 while much of association cortex has no accepted columnar parcellation. And
656 the column’s standing as a canonical cortical unit is contested (Horton &
657 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
658 supplies the oscillator index; the content cover remains an open empirical
659 question, which the bridge assumption E78 records.

660 The spatial reduction is also testable. A row-sum aggregation at zero
661 diagonal tracks the uniform-coupling control over the tested spatial decay
662 range, while frequency heterogeneity displaces the operational threshold. The

663 fit window biases both estimates relative to $2D$; agreement with the control
 664 is therefore the supported conclusion. Supplemental Material reports the
 665 reduction, its numerical error and the distinct assumptions needed for the
 666 kernel-average identification in Eq. (5).

667 The field commitment requires physical evidence beyond a choice of math-
 668 ematical representation, and the available support comparisons do not supply
 669 it. A finite matrix spread over positive-mass cells preserves its total cou-
 670 pling and phase correlation exactly, while point-supported vanishing under
 671 an atomless measure and a disadvantage from excluding a better-correlated
 672 pair concern specified support constraints; Section 6 reads the same compar-
 673 isons against a digital candidate. Neither supplies energy prices.

674 A spike-time shift alone supplies no coupling rate. The dimensionless
 675 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 676 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (12)$$

677 This rate describes how induced timing shifts accumulate into drift. Nei-
 678 ther it nor the installation factor κ follows from a heat bound. Testing the
 679 physical threshold requires independent estimates of the interaction kernel,
 680 its aggregation and diffusion in the same cortical regime.

681 9.2 Extracellular geometry and recovery

682 The proposed control variable is extracellular geometry. Extracellular vol-
683 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
684 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
685 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
686 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
687 an effective-medium description predicts that a larger extracellular fraction
688 raises conductivity and reduces field amplitude.

689 The coupling-gated hypothesis consequently predicts that extracellular
690 contraction on awakening raises E and, holding the other factors in Eq. (12)
691 fixed, raises effective coupling. This argument supplies a direction; the
692 geometry–coupling relation, its magnitude and its role in human recovery
693 require measurement.

694 For a smooth crossing at nonzero speed, evaluating the stationary branch
695 along the changing control gives a deterministic tracking approximation that
696 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (13)$$

697 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
698 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
699 proximation under the tested initialization. A separation of geometry and
700 phase time scales can support tracking away from threshold. The awakening

701 hypothesis needs a calibrated dynamical prediction for an operational onset
702 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
703 recovery latency.

704 **9.3 A protocol for the awakening experiment**

705 Neural inertia describes a behavioural barrier and hysteresis between anaes-
706 thetic induction and emergence (Friedman et al., 2010); recovery can pass
707 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
708 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
709 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
710 mechanism from a synaptic one. A possible discriminator is spatial onset
711 relative to independently estimated local field coupling and network connec-
712 tivity. The present mean-field model has no spatially varying κ and derives
713 no nucleation order, so a field-based onset map needs a separate model spec-
714 ified before the observations are inspected.

715 For that comparison, partition the recording into prespecified regions i
716 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
717 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
718 and a connectivity score c_i from a separately estimated structural network.
719 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
720 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

721 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (14)$$

722 Predicted onset is the first crossing of a region-specific threshold fixed from
723 pre-onset fluctuations. Fit the two scales on training sessions, then com-
724 pare held-out onset times and site order under the same observation model.
725 Larger field scores predict earlier onset only when baseline order and diffu-
726 sion do not reverse the ranking. This declared approximation omits interre-
727 gional coupling and saturation; a successful ranking would support its field
728 covariate, not prove ephaptic causation. Supplemental Material specifies the
729 controls and failure conditions.

730 A within-subject protocol can manipulate emergence rate, with spatially
731 resolved recordings to compare the proposed field and connectivity models:

- 732 1. **Manipulate emergence rate.** Vary the propofol infusion taper
733 across sessions within the same subject, recording drug concentrations
734 or a validated pharmacokinetic proxy to quantify the manipulation.
- 735 2. **Measure behavioural recovery.** Record time-to-response to a stan-
736 dard stimulus as the independent behavioural endpoint, rather than
737 estimating ΔK directly.
- 738 3. **Resolve spatial onset.** Identify anatomical locations where coher-
739 ence first crosses the baseline-scaled threshold and score the two maps

740 of Eq. (14) on held-out sessions. The two covariate maps are fixed
741 before the outcome sessions are inspected.

742 **4. Match physical controls.** Confirm that the observed coherence-to-
743 amplitude relation holds under the specific local diffusion D and ob-
744 servation noise. The model’s quadratic-variation rate $2D$ applies to
745 the latent phase; a band-limited analytic phase is smooth below its
746 correlation time, and on a constructed field its squared-increment rate
747 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
748 fore requires an explicit observation model for filtering and sampling.

749 Predeclare the recording window, regional fluctuation thresholds, and a
750 shared timing-error model. Fit the two scales and that observation model
751 on training sessions only; on held-out sessions, score the observed site-onset
752 times and sites that never cross by the same regional predictive rule. Report
753 site order as a secondary outcome and behavioural time-to-response sepa-
754 rately. Sessions without a crossing are censored observations, not zero delays
755 or exclusions. Compare the two scores only where independently estimated
756 field and connectivity maps predict distinguishable onset patterns at the
757 recording resolution. A held-out field-model advantage would support that
758 specified spatial predictor; it would not identify an ephaptic cause, shared
759 content or experience. Supplemental Material gives the controls and pilot
760 quantities needed to judge whether the comparison is informative.

761 10 Discussion

762 10.1 Unity and a minimal self as interpretive motiva- 763 tion

764 We propose the state glued from compatible local descriptions as a formal
765 correlate of conscious unity, and its accurate internal reconstruction as a cor-
766 relate of a minimal self in the sense of a pre-reflective point of view on the
767 scene (Gallagher, 2000). In the cup example, this would mean one description
768 that preserves the cup’s relation to the perceiver as well as the scene. Com-
769 patibility permits distinct contents and uncertainty; the shared quantities
770 must be chosen from independent evidence about what each region encodes.
771 The reconstruction need not be a verbal or autobiographical account. Its pro-
772 posed relation to experienced perspective remains an interpretation. Neither
773 identification is among the eight hypotheses in Table 1, and satisfying them
774 would not measure experience. The unity proposal also lacks an operational
775 definition. The gluing theorem holds for whatever cover it is given, a disjoint
776 cover satisfies it trivially, and the cover of interest overlaps in content rather
777 than in space (Section 4); until a principle fixes that cover, the correlate is a
778 schema awaiting one.

779 **10.2 A comparison that could disconfirm the link to** 780 **content**

781 The framework predicts that conscious content tracks compatibility, not co-
782 herence. Present matched stimuli that are reported on some trials and not
783 on others, for example at masking or rivalry threshold, and fit decoders for a
784 shared stimulus quantity independently on two overlapping cortical regions.
785 Compare decoded agreement on the overlap between reported and unreported
786 trials after matching their phase coherence. The proposed link predicts higher
787 agreement on reported trials at equal coherence. Equal agreement, or agree-
788 ment that follows coherence and not report, counts against it. A self-model
789 comparison has the same form: reconstruction of perceiver-relative variables,
790 such as hand position relative to the target, from a declared local encoding
791 should be more accurate on reported trials. These are correlational tests of
792 the proposed correlates; report is itself an access measure, and a no-report
793 design would be needed to separate content from access.

794 **10.3 What a test of the full proposal must establish**

795 A test needs separate estimates of effective coupling and diffusion, shared
796 content on overlapping regions, and reconstruction from a specified encoding
797 at a declared fidelity. The cover, relevant state family and decoder must
798 be fixed without selecting them for agreement. An independent experiential
799 endpoint is needed to evaluate the interpretation; absence of a report alone

800 does not define absence of experience. A unified episode with incompati-
801 ble decoded descriptions would challenge the unity identification. A unified
802 episode lacking the proposed self-encoding would challenge the minimal-self
803 claim. Conversely, satisfying the mathematical conditions without such an
804 episode would challenge their sufficiency as correlates.

805 For the physical route, cortical actuator modes and prices, the geometry-
806 coupling relation and a dynamical onset model remain to be measured. The
807 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
808 diction. Controlled changes in coordination and independently decoded con-
809 tent agreement would address the representational connection. The synthetic
810 decoder exercise calibrates part of these measurements, and the EEG exercise
811 identifies what a pooled bipolar estimand cannot measure; neither identifies
812 a neural content code or tests experience.

813 **10.4 Explanatory scope**

814 Integrated-information, global-workspace and predictive-processing accounts
815 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
816 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
817 relations among coupling, local content and reconstruction. Its formal field
818 condition requires a continuous kernel, a normalized substrate, a specified
819 average coupling and noise, and a site-contribution bound. Those conditions
820 do not identify an electromagnetic substrate; the extracellular interpretation
821 needs the separate calibration of Section 9. The joint witness shows that

822 the formal hypotheses can hold together in a declared model. A cortical
823 realization and an experiential interpretation require additional evidence.

824 Four limitations bound every claim above. Synchrony is not itself an
825 experiential measure, and nothing here establishes a consciousness criterion:
826 the conditions are proposed correlates to be tested against report or another
827 independent endpoint. The present EEG analysis checks a measurement
828 pipeline and does not validate any experiential claim. The formal results are
829 conditional on declared encoders, covers and maps whose cortical counter-
830 parts are unmeasured. No principle selects the cover on which compatibility
831 is assessed, so the unity correlate is not yet operationally defined.

832 The remaining explanatory question is whether these relations occur in
833 neural activity and matter for experience. The framework identifies candidate
834 correlates and tests of their physical and representational premises. It does
835 not explain why satisfying them would be accompanied by experience.

836 11 Conclusions

837 Phase coherence, compatibility and self-reconstruction are separable condi-
838 tions. Coherence constrains shared content only through encoders shared on
839 overlaps and with a bound that grows as the square root of the population
840 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch
841 order, so for populations of cortical size the positive link from coherence
842 to content is the bridge assumption E78 and not a derived result. Com-

843 patible local descriptions, not synchrony, determine one global state; and a
844 contracting self-map fixes that state while reconstructing only a family of
845 bounded diameter. Each result replaces an assumption about synchrony by
846 a measurement: independently decoded agreement on overlaps, a declared
847 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
848 ison of Section 10.2 states how such measurements could count against the
849 proposed link to conscious content.

850 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material's convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one

Table 2 (continued)

Symbol	Defined	Meaning
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§7, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (11)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (12)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d

Table 2 (continued)

Symbol	Defined	Meaning
E	Eq. (12)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

851 Data and code availability

852 The Lean development, simulation scripts, tests and saved numerical sum-
853 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
854 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
855 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication
856 simulation values are generated from saved summaries without rerunning
857 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
858 `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
859 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
860 and selection are described in Supplemental Material and implemented in
861 `simulations/empirical_collapse.py`. The repository contains the repro-
862 ducible analysis and does not redistribute the raw recordings.

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865 ing interests.

866 Use of AI tools

867 AI assistance to this work was substantial and is not separable by task.
868 Several coding assistants and chat interfaces were used in an interleaved way
869 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
870 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
871 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
872 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
873 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
874 they contributed to the Lean development, the simulation and analysis code,
875 and the drafting and revision of both documents. Because the tools were
876 used in combination and in alternation, no section, proof, figure or result is
877 attributable to any one of them, and none is claimed to be. The author is
878 responsible for all content, including everything a tool produced. Machine
879 checking establishes the stated Lean results under their hypotheses; it does
880 not validate biological interpretations or replace scientific review of generated
881 code and prose.

882 References

883 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
884 electric fields emerge from and allow representational drift. *NeuroImage*,
885 253, 119058.

- 886 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
887 tion theory: from consciousness to its physical substrate. *Nature Reviews*
888 *Neuroscience*, 17(7), 450-461.
- 889 Singer, W., & Gray, C. M. (1995). Visual feature integration and the temporal
890 correlation hypothesis. *Annual Review of Neuroscience*, 18, 555-586.
- 891 Fries, P. (2015). Rhythms for cognition: communication through coherence.
892 *Neuron*, 88(1), 220-235.
- 893 Shadlen, M. N., & Movshon, J. A. (1999). Synchrony unbound: a critical
894 evaluation of the temporal binding hypothesis. *Neuron*, 24(1), 67-77.
- 895 Gallagher, S. (2000). Philosophical conceptions of the self: implications for
896 cognitive science. *Trends in Cognitive Sciences*, 4(1), 14-21.
- 897 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
898 Mind. *Oxford University Press*.
- 899 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
900 tational theory of conscious processing. *Current Opinion in Neurobiology*,
901 25, 76-84.
- 902 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
903 *Reviews Neuroscience*, 11(2), 127-138.
- 904 Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness:

905 A case for biological computationalism. *Neuroscience & Biobehavioral Re-*
906 *views*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.

907 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
908 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
909 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.

910 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
911 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.

912 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
913 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
914 doi:10.1017/S0140525X26106906.

915 Gurnee, W., et al. (2026). Verbalizable Representations Form a
916 Global Workspace in Language Models. *Transformer Circuits Thread*.
917 arXiv:2607.15495.

918 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
919 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
920 40, 24-50.

921 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
922 parts of the free energy principle. *Entropy*, 23(3), 293.

923 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
924 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

925 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
926 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

927 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
928 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
929 *Progress in Neurobiology*, 226, 102465.

930 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
931 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
932 9(8), 45-60.

933 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
934 testable hypothesis about the characteristics of conscious as opposed to
935 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

936 McFadden, J. (2020). Integrating information in the brain's EM field: the
937 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
938 niaa016.

939 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
940 guide neocortical network activity. *Neuron*, 67(1), 129-143.

941 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
942 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
943 95-103.

944 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
945 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

946 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
947 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

948 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
949 *logical Reviews*, 98(1), 239-389.

950 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
951 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.

952 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
953 without a function. *Philosophical Transactions of the Royal Society B*,
954 360(1456), 837-862. doi:10.1098/rstb.2005.1623.

955 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
956 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.

957 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
958 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
959 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

960 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
961 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
962 ity of complexity measures to capture dreaming during propofol sedation.
963 *Scientific Reports*, 15, 32746.

964 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
965 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
966 brain. *Science*, 342(6156), 373-377.

- 967 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
968 *Physiological Reviews*, 88(4), 1277-1340.
- 969 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
970 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.
- 971 Logothetis, N. K., Kayser, C., & Oeltermann, A. (2007). In vivo measure-
972 ment of cortical impedance spectrum in monkeys: implications for signal
973 propagation. *Neuron*, 55(5), 809–823. doi:10.1016/j.neuron.2007.07.027.
- 974 Gabriel, S., Lau, R. W., & Gabriel, C. (1996). The dielectric properties of bio-
975 logical tissues: II. Measurements in the frequency range 10 Hz to 20 GHz.
976 *Physics in Medicine and Biology*, 41(11), 2251–2269. doi:10.1088/0031-
977 9155/41/11/002.
- 978 Norton, J. D. (2005). Eaters of the lotus: Landauer’s principle and the return
979 of Maxwell’s demon. *Studies in History and Philosophy of Science Part B:*
980 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- 981 Landauer, R. (1961). Irreversibility and heat generation in the computing
982 process. *IBM Journal of Research and Development*, 5(3), 183-191.
- 983 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
984 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 985 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
986 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

987 Strogatz, S. H., & Mirollo, R. E. (1991). Stability of incoherence in a popu-
988 lation of coupled oscillators. *Journal of Statistical Physics*, 63, 613–635.

989 Acebrón, J. A., Bonilla, L. L., Pérez Vicente, C. J., Ritort, F., &
990 Spigler, R. (2005). The Kuramoto model: A simple paradigm for
991 synchronization phenomena. *Reviews of Modern Physics*, 77, 137–185.
992 doi:10.1103/RevModPhys.77.137.

993 Ott, E., & Antonsen, T. M. (2008). Low dimensional behavior of
994 large systems of globally coupled oscillators. *Chaos*, 18, 037113.
995 doi:10.1063/1.2930766.

996 Dörfler, F., & Bullo, F. (2014). Synchronization in complex net-
997 works of phase oscillators: A survey. *Automatica*, 50, 1539–1564.
998 doi:10.1016/j.automatica.2014.04.012.

999 Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature*
1000 *Neuroscience*, 20, 340–352. doi:10.1038/nn.4497.

1001 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1002 of prediction. *Physical Review Letters*, 109(12), 120604.

1003 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1004 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1005 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1006 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1007 031015.

- 1008 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1009 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1010 PhysRevLett.111.180603.
- 1011 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1012 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 1013 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1014 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1015 181.
- 1016 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1017 archive.pitt.edu/115.
- 1018 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1019 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1020 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1021 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1022 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1023 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1024 Dai Pra, P., & den Hollander, F. (1996). McKean–Vlasov limit for interacting
1025 random processes in random media. *Journal of Statistical Physics*, 84, 735–
1026 772. doi:10.1007/BF02179656.
- 1027 Bertini, L., Giacomin, G., & Pakdaman, K. (2010). Dynamical aspects of

mean field plane rotators and the Kuramoto model. *Journal of Statistical Physics*, 138, 270–290. doi:10.1007/s10955-009-9908-9.

Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork bifurcations with additive noise. *Probability Theory and Related Fields*, 122, 341–388. doi:10.1007/s004400100174.

Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13, 113036. doi:10.1088/1367-2630/13/11/113036.

Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin, P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11), 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave patterns underlie complex brain dynamics and are related to cognitive processing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-023-01626-5.

Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information and dissipation with partially observed fluctuating signals. *Physical Review Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsubara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-

1049 cell resolution functional networks during unconsciousness are segre-
 1050 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
 1051 doi:10.1016/j.celrep.2025.116902.

1052 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
 1053 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
 1054 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.

1055 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
 1056 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
 1057 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.

1058 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
 1059 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.

1060 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.

1061 Baer, S. M., Erneux, T., & Rinzel, J. (1989). The slow passage through a
 1062 Hopf bifurcation: delay, memory effects, and resonance. *SIAM Journal on*
 1063 *Applied Mathematics*, 49(1), 55–71. doi:10.1137/0149003.

1064 Friedman, E. B., Sun, Y., Moore, J. T., Hung, H.-T., Meng, Q. C., Perera, P.,
 1065 Joiner, W. J., Thomas, S. A., Eckenhoff, R. G., Sehgal, A., & Kelz, M. B.
 1066 (2010). A conserved behavioral state barrier impedes transitions between
 1067 anesthetic-induced unconsciousness and wakefulness: evidence for neural
 1068 inertia. *PLoS ONE*, 5(7), e11903. doi:10.1371/journal.pone.0011903.

- 1069 Hudson, A. E., Calderon, D. P., Pfaff, D. W., & Proekt, A. (2014). Recovery
1070 of consciousness is mediated by a network of discrete metastable activity
1071 states. *Proceedings of the National Academy of Sciences*, 111(25), 9283–
1072 9288. doi:10.1073/pnas.1408296111.
- 1073 Proekt, A., & Hudson, A. E. (2018). A stochastic basis for neural inertia
1074 in emergence from general anaesthesia. *British Journal of Anaesthesia*,
1075 121(1), 86–94. doi:10.1016/j.bja.2018.02.035.